

September 10, 2020

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Fall Biosolids
Pace Project No.: 50265709

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on August 26, 2020. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Indianapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,



Melanie Booms
melanie.booms@pacelabs.com
(616)975-4500
Project Manager

Enclosures



REPORT OF LABORATORY ANALYSIS

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CERTIFICATIONS

Project: Fall Biosolids

Pace Project No.: 50265709

Pace Analytical Services Indianapolis

7726 Moller Road, Indianapolis, IN 46268

Illinois Accreditation #: 200074

Indiana Drinking Water Laboratory #: C-49-06

Kansas/TNI Certification #: E-10177

Kentucky UST Agency Interest #: 80226

Kentucky WW Laboratory ID #: 98019

Michigan Drinking Water Laboratory #9050

Ohio VAP Certified Laboratory #: CL0065

Oklahoma Laboratory #: 9204

Texas Certification #: T104704355

West Virginia Certification #: 330

Wisconsin Laboratory #: 999788130

USDA Soil Permit #: P330-19-00257

REPORT OF LABORATORY ANALYSIS

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SAMPLE SUMMARY

Project: Fall Biosolids

Pace Project No.: 50265709

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50265709001	Digester #3	Solid	08/25/20 13:30	08/26/20 10:30

REPORT OF LABORATORY ANALYSIS

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SAMPLE ANALYTE COUNT

Project: Fall Biosolids

Pace Project No.: 50265709

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50265709001	Digester #3	EPA 6010	RAM	6	PASI-I
		EPA 6020	DMT	10	PASI-I
		EPA 7471	ILP	1	PASI-I
		SM 2540G	SKK	1	PASI-I
		SM 2540G	SKK	1	PASI-I
		SM 2540G	MMS	1	PASI-I
		EPA 9045	SCM	1	PASI-I
		EPA 351.2	DAC1	1	PASI-I
		EPA 353.2	DAC1	2	PASI-I
		SM 4500-NH3 G	DAC1	1	PASI-I
		SM 4500-CI-E	SKK	1	PASI-I

PASI-I = Pace Analytical Services - Indianapolis

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Fall Biosolids
Pace Project No.: 50265709

Sample: Digester #3 **Lab ID: 50265709001** Collected: 08/25/20 13:30 Received: 08/26/20 10:30 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
6010 MET ICP								
Analytical Method: EPA 6010 Preparation Method: EPA 3050								
Pace Analytical Services - Indianapolis								
Aluminum	6480	mg/kg	1830	1	09/02/20 13:42	09/04/20 10:57	7429-90-5	
Calcium	32000	mg/kg	1830	1	09/02/20 13:42	09/04/20 10:57	7440-70-2	
Magnesium	5250	mg/kg	1830	1	09/02/20 13:42	09/04/20 10:57	7439-95-4	
Phosphorus	50100	mg/kg	1830	1	09/02/20 13:42	09/04/20 10:57	7723-14-0	N2
Potassium	3160	mg/kg	1830	1	09/02/20 13:42	09/04/20 10:57	7440-09-7	
Sodium	5280	mg/kg	1830	1	09/02/20 13:42	09/04/20 10:57	7440-23-5	
6020 MET ICPMS								
Analytical Method: EPA 6020 Preparation Method: EPA 3050B								
Pace Analytical Services - Indianapolis								
Arsenic	10.2	mg/kg	0.86	1	08/31/20 07:40	09/01/20 04:10	7440-38-2	
Cadmium	1.3	mg/kg	0.43	1	08/31/20 07:40	09/01/20 04:10	7440-43-9	
Chromium	101	mg/kg	1.7	1	08/31/20 07:40	09/01/20 04:10	7440-47-3	
Copper	978	mg/kg	4.3	10	08/31/20 07:40	09/01/20 03:15	7440-50-8	
Lead	23.1	mg/kg	4.3	5	08/31/20 07:40	09/01/20 03:01	7439-92-1	
Molybdenum	20.9	mg/kg	0.86	1	08/31/20 07:40	09/01/20 04:10	7439-98-7	N2
Nickel	36.2	mg/kg	0.43	1	08/31/20 07:40	09/01/20 04:10	7440-02-0	
Selenium	4.9	mg/kg	0.86	1	08/31/20 07:40	09/01/20 04:10	7782-49-2	
Silver	3.2	mg/kg	0.43	1	08/31/20 07:40	09/01/20 04:10	7440-22-4	
Zinc	1090	mg/kg	42.8	10	08/31/20 07:40	09/01/20 03:15	7440-66-6	
7471 Mercury								
Analytical Method: EPA 7471 Preparation Method: EPA 7471								
Pace Analytical Services - Indianapolis								
Mercury	<8.8	mg/kg	8.8	1	09/02/20 10:31	09/03/20 09:26	7439-97-6	
Percent Moisture								
Analytical Method: SM 2540G								
Pace Analytical Services - Indianapolis								
Percent Moisture	97.7	%	0.10	1		08/28/20 11:22		N2
2540G Total Percent Solids								
Analytical Method: SM 2540G								
Pace Analytical Services - Indianapolis								
Total Solids	2.3	%	0.10	1		08/28/20 11:22		N2
2540G Total Volatile Solids								
Analytical Method: SM 2540G								
Pace Analytical Services - Indianapolis								
Total Volatile Solids	52.7	%	0.10	1		08/28/20 11:22		N2
9045 pH Soil								
Analytical Method: EPA 9045								
Pace Analytical Services - Indianapolis								
pH at 25 Degrees C	8.3	Std. Units	0.10	1		08/31/20 13:28		H3
351.2 Total Kjeldahl Nitrogen								
Analytical Method: EPA 351.2 Preparation Method: EPA 351.2								
Pace Analytical Services - Indianapolis								
Nitrogen, Kjeldahl, Total	85900	mg/kg	10700	5	08/28/20 09:06	08/29/20 11:24	7727-37-9	N2

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Fall Biosolids

Pace Project No.: 50265709

Sample: Digester #3 **Lab ID: 50265709001** Collected: 08/25/20 13:30 Received: 08/26/20 10:30 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
353.2 Nitrogen, NO2/NO3								
Analytical Method: EPA 353.2 Preparation Method: EPA 353.2								
Pace Analytical Services - Indianapolis								
Nitrogen, NO2 plus NO3	<214	mg/kg	214	1	09/02/20 11:20	09/02/20 16:42		N2
Nitrogen, Nitrate	<214	mg/kg	214	1	09/02/20 11:20	09/02/20 16:42	14797-55-8	N2
4500 Ammonia Soil, Distilled								
Analytical Method: SM 4500-NH3 G Preparation Method: SM 4500-NH3 B								
Pace Analytical Services - Indianapolis								
Nitrogen, Ammonia	41300	mg/kg	2100	10	09/09/20 11:52	09/09/20 14:15	7664-41-7	N2
4500 Chloride in Soil								
Analytical Method: SM 4500-Cl-E Preparation Method: SM 4500-Cl-E								
Pace Analytical Services - Indianapolis								
Chloride	7310	mg/kg	4290	1	08/31/20 12:36	09/02/20 09:58	16887-00-6	N2

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50265709

QC Batch: 579516

Analysis Method: EPA 7471

QC Batch Method: EPA 7471

Analysis Description: 7471 Mercury

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50265709001

METHOD BLANK: 2671786

Matrix: Solid

Associated Lab Samples: 50265709001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Mercury	mg/kg	<0.20	0.20	09/03/20 08:20	

LABORATORY CONTROL SAMPLE: 2671787

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/kg	0.5	0.49	98	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2671788 2671789

Parameter	Units	50265807001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/kg	ND	0.49	0.52	0.53	0.54	106	105	75-125	2	20	

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QUALITY CONTROL DATA

Project: Fall Biosolids
Pace Project No.: 50265709

QC Batch: 579074	Analysis Method: EPA 6010
QC Batch Method: EPA 3050	Analysis Description: 6010 MET
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50265709001

METHOD BLANK: 2669906 Matrix: Solid

Associated Lab Samples: 50265709001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Aluminum	mg/kg	<50.0	50.0	09/04/20 10:55	
Calcium	mg/kg	<50.0	50.0	09/04/20 10:55	
Magnesium	mg/kg	<50.0	50.0	09/04/20 10:55	
Phosphorus	mg/kg	ND	50.0	09/04/20 10:55	N2
Potassium	mg/kg	<50.0	50.0	09/04/20 10:55	
Sodium	mg/kg	<50.0	50.0	09/04/20 10:55	

LABORATORY CONTROL SAMPLE: 2669907

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Aluminum	mg/kg	500	494	99	80-120	
Calcium	mg/kg	500	524	105	80-120	
Magnesium	mg/kg	500	490	98	80-120	
Phosphorus	mg/kg	50	53.6	107	80-120	N2
Potassium	mg/kg	500	488	98	80-120	
Sodium	mg/kg	500	500	100	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2669908 2669909

Parameter	Units	50265885003 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Aluminum	mg/kg	5910	485	486	8930	7390	623	303	75-125	19	20	P6
Calcium	mg/kg	12500	485	486	12400	32000	-40	4010	75-125	89	20	1d,P6
Magnesium	mg/kg	4470	485	486	5680	16800	249	2540	75-125	99	20	1d,P6
Phosphorus	mg/kg	427	48.5	48.6	563	452	282	51	75-125	22	20	1d,N2,P6
Potassium	mg/kg	768	485	486	1620	1310	175	111	75-125	21	20	1d,M0
Sodium	mg/kg	ND	485	486	537	490	99	89	75-125	9	20	

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QUALITY CONTROL DATA

Project: Fall Biosolids
Pace Project No.: 50265709

QC Batch: 579077	Analysis Method: EPA 6020
QC Batch Method: EPA 3050B	Analysis Description: 6020 MET
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50265709001

METHOD BLANK: 2669915 Matrix: Solid

Associated Lab Samples: 50265709001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/kg	<0.098	0.098	09/01/20 02:20	
Cadmium	mg/kg	<0.049	0.049	09/01/20 02:20	
Chromium	mg/kg	<0.20	0.20	09/01/20 02:20	
Copper	mg/kg	<0.049	0.049	09/01/20 02:20	
Lead	mg/kg	<0.098	0.098	09/01/20 02:20	
Molybdenum	mg/kg	<0.098	0.098	09/01/20 02:20	N2
Nickel	mg/kg	<0.049	0.049	09/01/20 02:20	
Selenium	mg/kg	<0.098	0.098	09/01/20 02:20	
Silver	mg/kg	<0.049	0.049	09/01/20 02:20	
Zinc	mg/kg	<0.49	0.49	09/01/20 02:20	

LABORATORY CONTROL SAMPLE: 2669916

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/kg	3.9	3.5	89	80-120	
Cadmium	mg/kg	3.9	3.9	99	80-120	
Chromium	mg/kg	3.9	3.9	100	80-120	
Copper	mg/kg	3.9	3.9	101	80-120	
Lead	mg/kg	3.9	4.0	104	80-120	
Molybdenum	mg/kg	3.9	3.9	100	80-120	N2
Nickel	mg/kg	3.9	3.9	101	80-120	
Selenium	mg/kg	3.9	3.5	89	80-120	
Silver	mg/kg	3.9	4.1	104	80-120	
Zinc	mg/kg	3.9	3.7	94	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2669917 2669918

Parameter	Units	50265901001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/kg	<10700 ug/kg	4.2	4.1	6.3	8.6	51	108	75-125	30	20	M3,R1
Cadmium	mg/kg	<215 ug/kg	4.2	4.1	4.1	3.9	94	94	75-125	3	20	
Chromium	mg/kg	13200 ug/kg	4.2	4.1	19.3	27.7	146	355	75-125	36	20	M3,R1
Copper	mg/kg	9260 ug/kg	4.2	4.1	11.7	16.0	58	166	75-125	31	20	M3,R1
Lead	mg/kg	13200 ug/kg	4.2	4.1	16.9	22.8	89	236	75-125	30	20	E,M0, R1

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50265709

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2669917 2669918												
Parameter	Units	50265901001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Molybdenum	mg/kg	<5360 ug/kg	4.2	4.1	4.6	5.1	76	92	75-125	12	20	N2
Nickel	mg/kg	12300 ug/kg	4.2	4.1	14.7	18.5	57	151	75-125	23	20	M3,R1
Selenium	mg/kg	<1070 ug/kg	4.2	4.1	4.2	4.0	83	81	75-125	5	20	
Silver	mg/kg	<107 ug/kg	4.2	4.1	4.0	3.9	95	94	75-125	4	20	
Zinc	mg/kg	31600 ug/kg	4.2	4.1	33.5	44.6	44	317	75-125	28	20	P6,R1

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50265709

QC Batch: 579085

Analysis Method: SM 2540G

QC Batch Method: SM 2540G

Analysis Description: Dry Weight/Percent Moisture

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50265709001

SAMPLE DUPLICATE: 2669937

Parameter	Units	50265709001 Result	Dup Result	RPD	Max RPD	Qualifiers
Percent Moisture	%	97.7	97.7	0	5	N2

SAMPLE DUPLICATE: 2669938

Parameter	Units	50265871007 Result	Dup Result	RPD	Max RPD	Qualifiers
Percent Moisture	%	10.2	10.9	6	5	N2,R1

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50265709

QC Batch: 579088

QC Batch Method: SM 2540G

Analysis Method: SM 2540G

Analysis Description: 2540G Total Solids

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50265709001

SAMPLE DUPLICATE: 2669945

Parameter	Units	50265709001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Solids	%	2.3	2.3	0	10	N2

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50265709

QC Batch: 579089

QC Batch Method: SM 2540G

Analysis Method: SM 2540G

Analysis Description: 2540G Total Volatile Solids

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50265709001

SAMPLE DUPLICATE: 2669946

Parameter	Units	50265709001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Volatile Solids	%	52.7	53.6	2	10	N2

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50265709

QC Batch: 579286

QC Batch Method: EPA 9045

Analysis Method: EPA 9045

Analysis Description: 9045 pH

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50265709001

SAMPLE DUPLICATE: 2671046

Parameter	Units	50265527003 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	7.7	7.7	0	2	H3

SAMPLE DUPLICATE: 2671047

Parameter	Units	50265965001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	6.2	6.2	0	2	H3

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50265709

QC Batch: 579056

Analysis Method: EPA 351.2

QC Batch Method: EPA 351.2

Analysis Description: 351.2 TKN

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50265709001

METHOD BLANK: 2669863

Matrix: Solid

Associated Lab Samples: 50265709001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	<49.3	49.3	08/29/20 11:23	N2

LABORATORY CONTROL SAMPLE: 2669864

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	490	515	105	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2669865 2669866

Parameter	Units	50265709001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Kjeldahl, Total	mg/kg	85900	21300	21100	106000	104000	93	85	90-110	2	20	M0, N2

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QUALITY CONTROL DATA

Project: Fall Biosolids
Pace Project No.: 50265709

QC Batch:	579798	Analysis Method:	EPA 353.2
QC Batch Method:	EPA 353.2	Analysis Description:	353.2 Nitrate + Nitrite
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50265709001

METHOD BLANK: 2673092 Matrix: Solid

Associated Lab Samples: 50265709001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Nitrate	mg/kg	<5.0	5.0	09/02/20 16:40	N2
Nitrogen, NO2 plus NO3	mg/kg	<5.0	5.0	09/02/20 16:40	N2

LABORATORY CONTROL SAMPLE: 2673093

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Nitrate	mg/kg	10	10.8	108	90-110	N2
Nitrogen, NO2 plus NO3	mg/kg	19.9	21.1	106	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2673094 2673095

Parameter	Units	50265709001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Nitrate	mg/kg	<214	431	431	456	475	104	108	90-110	4	20	N2
Nitrogen, NO2 plus NO3	mg/kg	<214	857	857	919	938	104	106	90-110	2	20	N2

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50265709

QC Batch: 580611

Analysis Method: SM 4500-NH3 G

QC Batch Method: SM 4500-NH3 B

Analysis Description: 4500 Ammonia, Distilled

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50265709001

METHOD BLANK: 2677114

Matrix: Solid

Associated Lab Samples: 50265709001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Ammonia	mg/kg	<5.0	5.0	09/09/20 14:13	N2

LABORATORY CONTROL SAMPLE: 2677115

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Ammonia	mg/kg	19.6	20.5	104	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2677116 2677117

Parameter	Units	50265709001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Ammonia	mg/kg	41300	814	831	42500	43500	149	268	90-110	2	20	N2,P6

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 50265709

QC Batch: 579356

Analysis Method: SM 4500-Cl-E

QC Batch Method: SM 4500-Cl-E

Analysis Description: 4500 Chloride

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50265709001

METHOD BLANK: 2671187

Matrix: Solid

Associated Lab Samples: 50265709001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Chloride	mg/kg	<99.9	99.9	09/02/20 09:39	N2

LABORATORY CONTROL SAMPLE: 2671188

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chloride	mg/kg	200	205	102	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2671189 2671190

Parameter	Units	50265702008 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Chloride	mg/kg	365	257	257	637	639	106	107	90-110	0	20	N2

MATRIX SPIKE SAMPLE: 2671191

Parameter	Units	50265718010 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Chloride	mg/kg	950	254	1180	92	90-110	N2

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

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QUALIFIERS

Project: Fall Biosolids
Pace Project No.: 50265709

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.
 ND - Not Detected at or above adjusted reporting limit.
 TNTC - Too Numerous To Count
 J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.
 MDL - Adjusted Method Detection Limit.
 PQL - Practical Quantitation Limit.
 RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.
 S - Surrogate
 1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.
 Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.
 LCS(D) - Laboratory Control Sample (Duplicate)
 MS(D) - Matrix Spike (Duplicate)
 DUP - Sample Duplicate
 RPD - Relative Percent Difference
 NC - Not Calculable.
 SG - Silica Gel - Clean-Up
 U - Indicates the compound was analyzed for, but not detected.
 N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.
 Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.
 TNI - The NELAC Institute.

ANALYTE QUALIFIERS

1d	RPD is outside control limits due to sample non-homogeneity RAM 9/4/20
E	Analyte concentration exceeded the calibration range. The reported result is estimated.
H3	Sample was received or analysis requested beyond the recognized method holding time.
M0	Matrix spike recovery and/or matrix spike duplicate recovery was outside laboratory control limits.
M3	Matrix spike recovery was outside laboratory control limits due to matrix interferences.
N2	The lab does not hold NELAC/TNI accreditation for this parameter but other accreditations/certifications may apply. A complete list of accreditations/certifications is available upon request.
P6	Matrix spike recovery was outside laboratory control limits due to a parent sample concentration notably higher than the spike level.
R1	RPD value was outside control limits.

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Fall Biosolids

Pace Project No.: 50265709

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50265709001	Digester #3	EPA 3050	579074	EPA 6010	580234
50265709001	Digester #3	EPA 3050B	579077	EPA 6020	579453
50265709001	Digester #3	EPA 7471	579516	EPA 7471	579942
50265709001	Digester #3	SM 2540G	579085		
50265709001	Digester #3	SM 2540G	579088		
50265709001	Digester #3	SM 2540G	579089		
50265709001	Digester #3	EPA 9045	579286		
50265709001	Digester #3	EPA 351.2	579056	EPA 351.2	579235
50265709001	Digester #3	EPA 353.2	579798	EPA 353.2	579816
50265709001	Digester #3	SM 4500-NH3 B	580611	SM 4500-NH3 G	580700
50265709001	Digester #3	SM 4500-Cl-E	579356	SM 4500-Cl-E	579759

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Sample Conditions Upon Receipt Form (SCUR)

Date/Time: 8/26/20 11:20		Evaluated by: AWS		WO#: 50265709 PM: MSB Due Date: 09/10/20 CLIENT: GR-Cadillac	
Client: CITY OF CADILLAC					
Project Manager: MSB		Profile ID:			
Rush TAT Requested: YES ☒ NO ☐ Due Date:					
Lab Notified of Rush or Short Holds: YES ☐ NO ☒		Non Conformance Form Required: YES ☐ NO ☒			
Samples Received Via: FedEx ☐ UPS ☒ Client ☐ Pace Courier ☐ Other: _____					
Comments:					
Custody Seals Present and Intact:		YES	NO	☒ N/A	
Received Sample Information Form(s): Drinking Waters Only		YES	NO	☒ N/A	
USDA Regulated Soils: (AL, AR, CA, FL, GA, ID, LA, MS, NM, NY, NC, OK, OR, SC, TN, TX, WA or Puerto Rico)		YES	☒ NO	☒ N/A AWS	
Short Holds Present (< 72 Hours):		YES	☒ NO		
Samples Received in Hold:		☒ YES	NO		
Custody Signatures Present:		☒ YES	NO		
Collector Signature Present:		☒ YES	NO		
Packing Material Used:		☒ YES	NO		
Samples Collected Today and On Ice:		YES	NO	☒ N/A	
IR Gun #: ☒ 280 281		Digital Thermometer #: 282 283			
Ice Type: WET Bagged / WET ☒ Loose BLUE NONE		1. Cooler Temp Upon Receipt: 3.0/3.5 °C			
Ice Location: TOP BOTTOM MIDDLE ☒ DISPERSED		Temp should be 0-6°C (Initial/Corrected)			
Temp Blank Received:		YES	☒ NO		
Containers Intact:		☒ YES	NO		
Correct Containers:		☒ YES	NO		
Sufficient Volume:		☒ YES	NO		
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation Exceptions are VOA, coliform, LLHg, O&G, or any container with a septum cap or preserved with HCl		YES	NO	☒ N/A	
Residual Chlorine Absent: (SVOC/Pest 625, PCB 608, Total/Amenable/Available Cyanide)		YES	NO	☒ N/A	
VOA Headspace Acceptable (<6mm):		YES	NO	☒ N/A	
Trip Blank Received: HCl MeOH TSP OTHER		YES	☒ NO		
Comments: N/A		2. Cooler Temp Upon Receipt: _____ °C			
		3. Cooler Temp Upon Receipt: _____ °C			
		4. Cooler Temp Upon Receipt: _____ °C			